ABSTRACT

The **BakeFlow – Bakery Production & Counter Management System** is a web-based application developed using React.js to improve and streamline the internal operations of a bakery. Traditional bakery operations often involve manual order handling, difficulty in monitoring product availability, inefficient coordination between counter staff and production staff, and limited visibility into daily production, sales, and inventory. The proposed system addresses these challenges by providing a centralized digital platform for managing bakery orders and internal operations efficiently.

The system consists of three primary user roles: **Counter Staff, Production Staff, and Administrator**. Counter staff can create and manage incoming orders, select bakery items and quantities, view product availability, and update order details. Production staff can view incoming orders, prepare the required bakery items, update production status, and mark orders as ready for collection. The administrator has control over the overall management operations, including product and price management, inventory management, staff management, order monitoring, sales tracking, and generation of basic reports.

The application is designed with a role-based access structure so that each staff member can access only the features relevant to their responsibilities. The internal management component forms the core of the system, allowing bakery staff and administrators to monitor and manage daily operations efficiently. Digital order and production status information helps reduce confusion, avoid delays, prevent orders for unavailable products, and improve coordination between the counter and production sections.

The frontend of the system is developed using **React.js**, providing a responsive and interactive user interface with reusable components and efficient state management. The system can be integrated with a suitable backend and database for secure storage and management of staff information, orders, products, inventory, production records, and sales data. The proposed system focuses on simplifying the essential bakery operations through a clear digital workflow such as **New Order, Preparing, Ready, and Completed**, without including unnecessary customer-side features. Overall, BakeFlow aims to provide a centralized, efficient, and user-friendly solution for managing bakery operations while reducing manual work and supporting better coordination and decision-making among bakery staff.